Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction

Period descriptors, PP-kernel nodes,
and the full tractability of RCC5

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Claude (Anthropic)
Prompted by Michael Wessel

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Disclaimer. This document was generated by Claude (Anthropic), prompted by Michael Wessel. The results have not been independently verified. We invite scrutiny and corrections.

Revision history. This is the fourth revision. The first version was reviewed by GPT-5.4 Pro [5], who raised five objections; these were addressed in the second version [3]. GPT-5.4's second review [6] raised four further objections, addressed in the third version. GPT-5.4's third review [7] raised four more objections. This fourth version incorporates all fixes: a new phasewise safety validity condition (V_{safe}); exact-relation extraction proofs via backward forcing (DR, PP) and forward absorption (PPI); a constructive quotient approach that breaks the size-bound circularity; and an honest acknowledgment of a remaining PO gap.

Abstract

We study the decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$—the description logic $\mathcal{ALCI}$ extended with an RCC5 role box. We introduce a finite *two-tier quotient*: Tier 1 captures each infinite PP-chain's eventually periodic tail as a *period descriptor*, while Tier 2 captures cross-chain interactions through *PP-kernel nodes*. A valid quotient can be unfolded into a genuine model using the *full tractability* of RCC5 (Renz & Nebel 1999).

For demands involving DR, PP, and PPI, we prove complete decidability. The key ingredients are: (i) a *backward forcing* property for DR and PP (the composition table forces these relations backward along PP-chains, guaranteeing all-phase type-safety of witness types); (ii) *forward absorption* for PPI ($\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ ensures stabilized witnesses); (iii) a *constructive quotient* with one kernel per descriptor, giving a computable size bound with no circularity; and (iv) a *chain-unfolding lift lemma* bridging the finite quotient to the infinite model.

For PO, a genuine gap remains: the composition table provides neither backward forcing nor forward absorption, and we exhibit a realizable descriptor where the constant-interface architecture is provably incomplete. We precisely characterize the *PO-coherent fragment* where the proof succeeds and identify the PO obstacle as the key remaining open problem.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with five jointly exhaustive and pairwise disjoint (JEPD) base relations from the Region Connection Calculus RCC5 [12]: DR, PO, EQ, PP, PPI.

Wessel [1] introduced this logic and left its decidability open. The logic lacks both the finite model property and the tree model property: the concept $C_\infty = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$ is satisfiable only in infinite models containing unbounded PP-chains.

Several recent approaches have been attempted: a quasimodel method [2] (gap in completeness), a contextual tableau [4] (whose completeness conjecture $\mathsf{FW}(C, N)$ is false [8]), and a triangle-type approach [9] (conditional on an unproven conjecture). All share a common difficulty: assigning globally consistent RCC5 edges in complete-graph models.

Results. We prove decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ for the *PO-coherent fragment* (Definition 5.4), which includes all concepts without $\forall \mathsf{PO}$ restrictions and all concepts where the $\forall \mathsf{PO}$ restrictions are uniform across phases. For the full logic, we identify a precisely characterized PO obstacle (§10) and leave its resolution as the main open problem.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts are built from concept names, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$ in negation normal form (NNF). We adopt **strong EQ semantics**.

An interpretation $\mathcal{I} = (\Delta^\mathcal{I}, \cdot^\mathcal{I})$ assigns a total function $\rho^\mathcal{I} : \Delta^\mathcal{I} \times \Delta^\mathcal{I} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ subject to: (i) $\rho(d, d) = \mathsf{EQ}$; (ii) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$; (iii) composition consistency: $\rho(d_1, d_3) \in \mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$ for all d_1, d_2, d_3.

2.2 Types, closure, composition table

Given C_0 in NNF, $\mathsf{cl}(C_0)$ is the closure under subformulas. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset. Write $\mathsf{Tp}(C_0)$ for the set of types; $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

The RCC5 composition table $\mathsf{comp} : \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}^2 \to 2^{\{\mathsf{DR},\mathsf{PO},\mathsf{PP},\mathsf{PPI}\}}$ (with EQ excluded for distinct elements) is:

$\mathsf{comp}(R_1, R_2)$	DR	PO	PP	PPI
DR	{all}	{DR,PO,PP}	{DR,PO,PP}	{DR}
PO	{DR,PO,PPI}	{all}	{PO,PP}	{DR,PO,PPI}
PP	{DR}	{DR,PO,PP}	{PP}	{all}
PPI	{DR,PO,PPI}	{PO,PPI}	{PO,PP,PPI}	{PPI}

2.3 Type-safety for RCC5 relations

Definition 2.1 (Type-safe relations)**.** For Hintikka types $\tau_1, \tau_2 \in \mathsf{Tp}(C_0)$, define

$$\mathsf{Safe}(\tau_1, \tau_2) = \{R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} : R \text{ respects all universals in } \tau_1 \text{ and } \tau_2\}$$

where R *respects the universals* means: for every $\forall R.C \in \tau_1$, $C \in \tau_2$; and for every $\forall \mathsf{inv}(R).C \in \tau_2$, $C \in \tau_1$.

Lemma 2.2 (Model relations are type-safe)**.** *In any model $\mathcal{I}$, for distinct $d, e \in \Delta^\mathcal{I}$: $\rho(d, e) \in \mathsf{Safe}(\mathsf{tp}(d), \mathsf{tp}(e))$.*

Proof. Immediate from the semantics: if $\forall R.C \in \mathsf{tp}(d)$ and $\rho(d,e) = R$, then $e \in C^{\mathcal{I}}$, so $C \in \mathsf{tp}(e)$. The inverse direction is symmetric. □

2.4 Full tractability of RCC5

Theorem 2.3 (Full RCC5 tractability, Renz & Nebel [10], Renz [11])**.** *Every path-consistent disjunctive RCC5 constraint network over the base relations* $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ *is satisfiable. Equivalently: if path-consistency enforcement does not empty any domain, then a consistent atomic refinement exists.* □

This is the "patchwork property" of RCC5: local consistency (path-consistency) implies global consistency.

3 Algebraic properties of PP

Theorem 3.1 (Universal self-absorption of PP)**.** *(a)* ***Left absorption:*** $R \in \mathsf{comp}(\mathsf{PP}, R)$ *for every* $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

(b) ***Right absorption:*** $R \in \mathsf{comp}(R, \mathsf{PP})$ *for every* R.

(c) ***PPI right-absorption:*** $R \in \mathsf{comp}(\mathsf{PPI}, R)$ *for every* R.

(d) ***PPI left-absorption:*** $R \in \mathsf{comp}(R, \mathsf{PPI})$ *for every* R.

Proof. Direct verification against the composition table. □

Corollary 3.2 (Reflexive PP is composition-consistent)**.** *A node* k *with* $\rho(k,k) = \mathsf{PP}$ *satisfies all composition constraints with any external node* m. □

Theorem 3.3 (PP-transitivity)**.** $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$. *Consequently,* PP *on any finite set of elements is a strict partial order.* □

Lemma 3.4 (Backward forcing)**.** *Let* $(d_i)_{i \geq 0}$ *be a* PP*-chain with* $\rho(d_j, w) = R$ *for some element* w *and position* j.

(a) If $R = \mathsf{DR}$*: then* $\rho(d_i, w) = \mathsf{DR}$ *for all* $i \leq j$.

(b) If $R = \mathsf{PP}$*: then* $\rho(d_i, w) = \mathsf{PP}$ *for all* $i \leq j$.

Proof. For $i < j$: $\rho(d_i, d_j) = \mathsf{PP}$, so $\rho(d_i, w) \in \mathsf{comp}(\mathsf{PP}, \rho(d_j, w)) = \mathsf{comp}(\mathsf{PP}, R)$. Now $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$: both are singletons forcing the same value backward. □

Remark 3.5 (Why backward forcing fails for PO and PPI)**.** $\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\text{all}\}$ are not singletons. Hence PO and PPI are *not* backward-forced: knowing $\rho(d_j, w) = \mathsf{PO}$ does not determine $\rho(d_i, w)$ for $i < j$. This asymmetry is the algebraic root of the PO gap identified in §10.

Lemma 3.6 (Forward transition table)**.** *Along a* PP*-chain* (d_i)*, the valid transitions for* $\rho(d_i, w) \to \rho(d_{i+1}, w)$ *(satisfying* both *the forward constraint* $\rho(d_{i+1}, w) \in \mathsf{comp}(\mathsf{PPI}, \rho(d_i, w))$ *and the backward constraint* $\rho(d_i, w) \in \mathsf{comp}(\mathsf{PP}, \rho(d_{i+1}, w))$*) are:*

Current	*Valid next values*
DR	DR, PO, PPI
PO	PO, PPI
PP	PO, PP, PPI
PPI	PPI

In particular: (i) PPI *is forward-absorbing; (ii) once the sequence leaves* DR *(to* PO *or* PPI*), it never returns to* DR*; (iii) once it leaves* PO *(to* PPI*), it never returns to* PO*.*

Proof. The forward constraint gives: $\mathsf{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$. The backward constraint eliminates further values. For instance, from DR: the forward constraint allows $\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}$; checking $\mathsf{DR} \in \mathsf{comp}(\mathsf{PP}, R')$ for each candidate R': $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ ($\mathsf{DR} \in \checkmark$), $\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ ($\mathsf{DR} \in \checkmark$), $\mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\text{all}\}$ ($\mathsf{DR} \in \checkmark$). All survive. From PO: forward allows $\mathsf{PO}, \mathsf{PPI}$; backward check: $\mathsf{PO} \in \mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ ($\checkmark$), $\mathsf{PO} \in \mathsf{comp}(\mathsf{PP}, \mathsf{PPI}) = \{\text{all}\}$ ($\checkmark$). Both survive. The no-return properties follow by inspection. □

4 PP-chain structure

Definition 4.1 (PP-chain)**.** In a model $\mathcal{I}$, a PP-*chain* is a sequence $(d_i)_{i\geq 0}$ with $\rho(d_i, d_{i+1}) = \mathsf{PP}$ for all i. By Theorem 3.3, $\rho(d_i, d_j) = \mathsf{PP}$ for $i < j$.

Lemma 4.2 (External relation stabilization)**.** *For any element e and chain (d_i), the sequence $\rho(d_i, e)$ eventually stabilizes to a constant value.*

Proof. By Lemma 3.6, the sequence takes values in $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ with only the listed transitions. The non-return properties (once leaving DR, PO, or PP, the sequence cannot return) ensure eventual constancy. □

Definition 4.3 (Recurrent types, T_∞)**.** For a PP-chain $(d_i)_{i\geq 0}$, define $T_\infty = \{\tau \in \mathsf{Tp}(C_0) : \tau = \mathsf{tp}(d_i) \text{ for infinitely many } i\}$.

Lemma 4.4 (T_∞ is non-empty and eventually exhaustive)**.** *$T_\infty \neq \emptyset$. There exists M such that $\mathsf{tp}(d_i) \in T_\infty$ for all $i \geq M$.*

Proof. Since $|\mathsf{Tp}(C_0)|$ is finite and the sequence $(\mathsf{tp}(d_i))_{i\geq 0}$ is infinite, some type appears infinitely often (pigeonhole), so $T_\infty \neq \emptyset$. Types not in T_∞ appear only finitely often; let M be the last index at which any such type appears. □

Lemma 4.5 (Phase-uniform type-safety)**.** *Let (d_i) be a* PP*-chain in a model $\mathcal{I} \models C_0$ with $T_\infty = \{\tau_1, \ldots, \tau_p\}$. Let w be any element with stabilized relation $S = \rho(d_i, w)$ for all sufficiently large i. Then $S \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$ for every $\tau_j \in T_\infty$.*

Proof. Fix $\tau_j \in T_\infty$. Choose k large enough that both $\mathsf{tp}(d_k) = \tau_j$ and $\rho(d_k, w) = S$. For any $\forall S.C \in \tau_j$: the model satisfies $\forall S.C$ at d_k, and $\rho(d_k, w) = S$, so $C \in \mathsf{tp}(w)$. The inverse direction is symmetric. Hence $S \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$. □

Lemma 4.6 (Backward forcing gives all-phase safety)**.** *Let (d_i) be a* PP*-chain, $T_\infty = \{\tau_1, \ldots, \tau_p\}$, and $R \in \{\mathsf{DR}, \mathsf{PP}\}$. If $\rho(d_j, w) = R$ for some position $j > M$ (past the exhaustion index), then $R \in \mathsf{Safe}(\tau_k, \mathsf{tp}(w))$ for every $\tau_k \in T_\infty$.*

Proof. By Lemma 3.4, $\rho(d_i, w) = R$ for all $i \leq j$. Since $j > M$, every $\tau_k \in T_\infty$ appears at some position $i \leq j$. At that position, $\rho(d_i, w) = R$ and $\mathsf{tp}(d_i) = \tau_k$, so $R \in \mathsf{Safe}(\tau_k, \mathsf{tp}(w))$ by Lemma 2.2. □

5 Period descriptors

Definition 5.1 (Period descriptor)**.** A *period descriptor* for C_0 is a tuple $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ with $p \geq 1$ and each $\tau_j \in \mathsf{Tp}(C_0)$, satisfying:

(V1) Each τ_j is a Hintikka type in $\mathsf{cl}(C_0)$.

(V2) **$\forall$PP-closure:** $\forall\mathsf{PP}.C \in \tau_i$ for some i implies $C \in \tau_j$ for all j.

(V3) **$\forall$PPI-closure:** $\forall\mathsf{PPI}.C \in \tau_i$ for some i implies $C \in \tau_j$ for all j.

(V4) **Chain continuation:** For each i, there exists $\exists\mathsf{PP}.C \in \tau_i$ such that $C \in \tau_j$ for some j.

Theorem 5.2 (T_∞ yields a valid period descriptor)**.** *Let $(d_i)_{i\geq 0}$ be a* PP*-chain in a model $\mathcal{I} \models C_0$. Any enumeration of $T_\infty = \{\tau_1, \ldots, \tau_p\}$ is a valid period descriptor.* □

Definition 5.3 (Stabilized core and union)**.** For $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$:

- $\mathsf{Core}(\mathsf{Desc}) = \bigcap_{j=1}^{p} \tau_j$.
- $\mathsf{Union}(\mathsf{Desc}) = \bigcup_{j=1}^{p} \tau_j$.

Definition 5.4 (PO-coherent descriptor)**.** A period descriptor $\mathsf{Desc} = (\tau_1, \ldots, \tau_p)$ is *PO-coherent* if for every $\exists\mathsf{PO}.C \in \mathsf{Union}(\mathsf{Desc})$, there exists a Hintikka type σ with $C \in \sigma$ and $\mathsf{PO} \in \bigcap_{j=1}^{p} \mathsf{Safe}(\tau_j, \sigma)$.

A concept C_0 is *PO-coherent* if every realizable period descriptor (i.e., one extractable as T_∞ from a model of C_0) is PO-coherent.

Remark 5.5 (When PO-coherence holds automatically)**.** A descriptor is automatically PO-coherent if: (a) it contains no $\exists\mathsf{PO}$ demands, or (b) for every $\forall\mathsf{PO}.D$ appearing in *any* phase, $D \in \mathsf{Core}(\mathsf{Desc})$ (so the $\forall\mathsf{PO}$ universals are the same across all phases), or (c) no phase contains any $\forall\mathsf{PO}$ restriction at all. The PO-incoherent case requires $\exists\mathsf{PO}.A$ in one phase and $\forall\mathsf{PO}.\neg A$ in a different phase—a relatively specific configuration.

6 The two-tier quotient

Definition 6.1 (Two-tier quotient)**.** A *two-tier quotient* $\mathcal{Q}$ for C_0 consists of:

Tier 1 (within-chain structure): A finite set of PO-coherent period descriptors $\{\mathsf{Desc}_1, \ldots, \mathsf{Desc}_K\}$, each satisfying (V1)–(V4).

Tier 2 (cross-chain structure):

(T1) A set of *PP-kernel nodes* $\{k_1, \ldots, k_K\}$, one per descriptor. Each k_α is associated with descriptor Desc_α. We set $\mathsf{tp}(k_\alpha) = \mathsf{Core}(\mathsf{Desc}_\alpha)$ and $\rho(k_\alpha, k_\alpha) = \mathsf{PP}$ (bookkeeping device; see Remark 6.2).

(T2) A set of *regular nodes* $\{m_1, \ldots, m_L\}$ with types $\mathsf{tp}(m_\beta) \in \mathsf{Tp}(C_0)$.

(T3) An atomic RCC5 labeling ρ on all pairs of kernel and regular nodes, satisfying composition consistency (V5) and phasewise type-safety (V_{safe}).

(T4) **Designated witnesses for kernel demands.** For each kernel node k_α and each non-PP existential demand $\exists R.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ with $R \neq \mathsf{PP}$: a designated witness w among the kernel/regular nodes, such that $C \in \mathsf{tp}(w)$, $\rho(k_\alpha, w) = R$, and

$$R \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w)) \quad \text{for } \textit{every} \text{ phase type } \tau_j \text{ of } \mathsf{Desc}_\alpha. \tag{1}$$

(T5) For each regular node m_β and each existential demand $\exists R.C \in \mathsf{tp}(m_\beta)$: a designated witness.

Validity conditions:

(V5) **Composition consistency:** The atomic network on kernel and regular nodes is composition-consistent (CC).

(V6) **Off-chain extension:** For each $\exists \mathsf{PP}.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ not satisfied within Desc_α (i.e., $C \notin \tau_j$ for any j), there is a witness node w with $\rho(k_\alpha, w) = \mathsf{PP}$, $C \in \mathsf{tp}(w)$, and $\mathsf{PP} \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$ for every phase type τ_j. The disjunctive constraint network obtained by adding all such witnesses simultaneously (with domains

$$D(w, e) = \mathsf{comp}(\text{demanded}, \rho(\text{parent}, e)) \ \cap \ \mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e)) \tag{2}$$

for all existing nodes e) is path-consistent after arc-consistency enforcement.

Phasewise type-safety:

(V_{safe}) For every kernel k_α and every external node e (kernel or regular):

$$\rho(k_\alpha, e) \ \in \bigcap_{\tau \in \mathsf{Desc}_\alpha} \mathsf{Safe}(\tau, \mathsf{tp}(e)).$$

For kernel–kernel edges, the condition applies on both sides.

Remark 6.2 (The quotient is a certificate, not a model)**.** The two-tier quotient is *not* itself an RCC5 network in the strong-EQ sense: kernel nodes have reflexive PP-loops. The quotient is an abstract certificate from which a genuine strong-EQ model can be constructed (Theorem 8.3). During chain unfolding, each kernel is replaced by infinitely many distinct elements with EQ self-loops.

Remark 6.3 (All-phase type-safety checks)**.** The type-safety conditions in T4 (1), V6, and V_{safe} are checked against *every* phase type τ_j of Desc_α. This is necessary because the soundness proof assigns the kernel's relation uniformly to all chain positions. Lemma 4.5 guarantees that model-derived relations always pass this all-phase check.

7 Completeness: model implies quotient

Theorem 7.1 (Constructive quotient extraction)**.** *If C_0 is satisfiable and PO-coherent, there exists a valid two-tier quotient $\mathcal{Q}$ for C_0 of size at most $f(|C_0|)$, where f is a computable function.*

Proof. Let $\mathcal{I} \models C_0$. Form the tree unraveling $\mathcal{I}^T$.

Step 1: One kernel per descriptor. For each maximal PP-chain in $\mathcal{I}^T$, compute T_∞. By Theorem 5.2, T_∞ is a valid period descriptor. Group chains by descriptor. Select one representative chain per distinct descriptor. This gives $K \leq D$ kernels, where $D \leq 2^{|\mathsf{Tp}(C_0)|} \leq 2^{2^{|\mathsf{cl}(C_0)|}}$ is a computable bound on the number of distinct descriptors.

Step 2: Kernel-to-kernel relations. For each pair of representative chains α, β: the doubly stabilized relation $\rho(k_\alpha, k_\beta)$ is $\lim_{i,j\to\infty} \rho(d_i^{(\alpha)}, d_j^{(\beta)})$. This is well-defined by Lemma 4.2 applied twice.

By Lemma 4.5, $\rho(k_\alpha, k_\beta) \in \mathsf{Safe}(\tau, \mathsf{Core}(\mathsf{Desc}_\beta))$ for every $\tau \in \mathsf{Desc}_\alpha$ (and symmetrically). So V_{safe} holds for kernel–kernel edges. Composition consistency follows from the model (choose sufficiently large indices).

Step 3: Designated witnesses. For each kernel k_α and each $\exists R.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ with $R \neq \mathsf{PP}$:

Case $R = \mathsf{DR}$: Choose a phase τ_j containing $\exists\mathsf{DR}.C$ and a position $i_n > M$ with $\mathsf{tp}(d_{i_n}) = \tau_j$. The model provides a DR-witness w with $\rho(d_{i_n}, w) = \mathsf{DR}$ and $C \in \mathsf{tp}(w)$. By Lemma 4.6, $\mathsf{DR} \in \mathsf{Safe}(\tau_k, \mathsf{tp}(w))$ for all $\tau_k \in T_\infty$. Create a quotient witness of type $\mathsf{tp}(w)$ with $\rho(k_\alpha, w') = \mathsf{DR}$. The all-phase check (1) passes.

Case $R = \mathsf{PPI}$: At a position i_n past the stabilization threshold for some witness w with $\rho(d_{i_n}, w) = \mathsf{PPI}$ and $C \in \mathsf{tp}(w)$: the forward absorption of PPI (Lemma 3.6) gives $S_w = \mathsf{PPI}$. By Lemma 4.5, all-phase safety holds.

Case $R = \mathsf{PO}$: By PO-coherence (Definition 5.4), there exists a type σ with $C \in \sigma$ and $\mathsf{PO} \in \bigcap_j \mathsf{Safe}(\tau_j, \sigma)$. The model guarantees that σ is a realized type (it appears as the type of a PO-witness at some chain position). Create a quotient witness of type σ with $\rho(k_\alpha, w') = \mathsf{PO}$.

The total number of designated witnesses is at most $K \cdot |\mathsf{cl}(C_0)|$ (one per demand per kernel), so $L \leq D \cdot |\mathsf{cl}(C_0)|$.

Step 4: Witness-to-other relations and V5. For each designated witness w' and each other quotient node e: the model provides a globally consistent relation $\rho(w, e)$ for the original witness element. By the AC-survival argument (below), these values survive arc-consistency enforcement.

The AC-survival argument: each off-chain witness w in the model has actual relations to all other elements. For any existing node e, $\rho(w, e) \in D(w, e)$ (the initial domain from (2)), because $\rho(w, e) \in \mathsf{comp}(\rho(w, \text{parent}), \rho(\text{parent}, e))$ (composition) and $\rho(w, e) \in \mathsf{Safe}(\mathsf{tp}(w), \mathsf{tp}(e))$ (model type-safety). Arc-consistency cannot remove $\rho(w, e)$ because every mediator e' satisfies $\rho(w, e') \in D(w, e')$ and $\rho(w, e') \in \mathsf{comp}(\rho(w, e), \rho(e, e'))$ (global consistency). So all domains remain non-empty.

Step 5: V_{safe} for kernel-to-regular edges. Each regular node inherits its model relations. By Lemma 4.5, the stabilized kernel-to-regular relation satisfies V_{safe}.

Step 6: Off-chain PP-witnesses (V6). For each $\exists\mathsf{PP}.C \in \mathsf{Union}(\mathsf{Desc}_\alpha)$ not satisfied within Desc_α: the model provides a PP-witness w at some chain position $i_n > M$. By Lemma 4.6 (with $R = \mathsf{PP}$), $\mathsf{PP} \in \mathsf{Safe}(\tau_j, \mathsf{tp}(w))$ for every phase τ_j. The AC-survival argument ensures the V6 path-consistency check passes.

Size bound. $K + L \leq D + D \cdot |\mathsf{cl}(C_0)| = D\,(1 + |\mathsf{cl}(C_0)|)$, where $D \leq 2^{2^{|\mathsf{cl}(C_0)|}}$. This is doubly exponential but **computable**, with no circular dependency. □

8 Soundness: quotient implies model

Lemma 8.1 (Chain-unfolding lift)**.** *Let $\mathcal{N}$ be a CC atomic RCC5 network on kernel and regular nodes. For each kernel k_α with descriptor $\mathsf{Desc}_\alpha = (\tau_1, \ldots, \tau_p)$, replace k_α by an infinite chain $d_1^{(\alpha)}, d_2^{(\alpha)}, \ldots$ with:*

- $\rho(d_i^{(\alpha)}, d_j^{(\alpha)}) = \mathsf{PP}$ *for* $i < j$, EQ *for* $i = j$*;*
- $\rho(d_i^{(\alpha)}, e) = \rho(k_\alpha, e)$ *for every non-chain node* e *and all* i*;*
- $\rho(d_i^{(\alpha)}, d_j^{(\beta)}) = \rho(k_\alpha, k_\beta)$ *for distinct chains* $\alpha \neq \beta$*.*

Then the unfolded network $\mathcal{N}'$ is CC.

Proof. Every triple in $\mathcal{N}'$ falls into one of six cases:

Case 1: within-chain (d_i, d_j, d_k), $i < j < k$. $\mathsf{PP} \in \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$. ✓

Case 2: two same-chain elements d_i, d_j $(i < j)$ and external e. $\rho(d_i, e) = \rho(d_j, e) = R$ (constant interface). Need: $R \in \mathsf{comp}(\mathsf{PPI}, R)$. Holds by Theorem 3.1(c). ✓

Case 3: one chain element $d_i^{(\alpha)}$ and two externals e_1, e_2. Reduces to triple (k_α, e_1, e_2), CC in $\mathcal{N}$. ✓

Case 4: elements from two chains and a third node. Reduces to (k_α, k_β, e), CC in $\mathcal{N}$. ✓

Case 5: three elements from three chains. Reduces to $(k_\alpha, k_\beta, k_\gamma)$, CC in $\mathcal{N}$. ✓

Case 6: two same-chain elements and one from another chain. Reduces to Case 2 with $e = d_j^{(\beta)}$. ✓ □

Corollary 8.2 (V6 lifts to the unfolded model)**.** *If the finite V6 extension network is path-consistent, then the unfolded network is also path-consistent.* □

Theorem 8.3 (Quotient realization)**.** *If a valid two-tier quotient $\mathcal{Q}$ exists for C_0, then C_0 is satisfiable.*

Proof. We construct a model $\mathcal{I}' \models C_0$.

Step 1: Unfold descriptors into chains. For each $\mathsf{Desc}_\alpha = (\tau_1, \ldots, \tau_p)$, create an infinite chain $d_1^{(\alpha)}, d_2^{(\alpha)}, \ldots$ with $\mathsf{tp}(d_i^{(\alpha)}) = \tau_{((i-1) \bmod p)+1}$ and $\rho(d_i, d_j) = \mathsf{PP}$ for $i < j$, EQ for $i = j$.

Step 2: Assign constant kernel interfaces. $\rho(d_i^{(\alpha)}, e) = \rho(k_\alpha, e)$ for every non-chain node e.

Step 3: CC of the unfolded base. By Lemma 8.1.

Step 4: Add off-chain witnesses. V6 ensures path-consistency. By Corollary 8.2, this lifts to the unfolded network. By Theorem 2.3, a consistent atomic refinement exists. Every value in the refined domains is in Safe, so the chosen atomic relations respect all universal constraints.

Step 5: Demand satisfaction.

- $\exists\mathsf{PP}$ **within chains:** If $C \in \tau_j$ for some j, the next occurrence of τ_j on the chain witnesses the demand. Otherwise, V6 provides an off-chain PP-witness.

- **Non-PP demands of chain elements:** For $\exists R.C \in \tau_i$ with $R \neq \mathsf{PP}$: the designated witness w (T4) has $\rho(k_\alpha, w) = R$, so $\rho(d_j^{(\alpha)}, w) = R$ (Step 2), witnessing the demand.
- **Universal constraints:** For $\forall R.C \in \tau_i$ and an external e with $\rho(d_j, e) = R$: V_{safe} guarantees $R \in \mathsf{Safe}(\tau_i, \mathsf{tp}(e))$, so $C \in \mathsf{tp}(e)$.

Step 6: Global consistency. The full atomic network is CC (Steps 3–4). By the patchwork property, this CC atomic network is globally consistent and realizable, giving $\mathcal{I}' \models C_0$. □

9 Main result

Theorem 9.1 (Decidability of $\mathcal{ALCI}_{RCC5}$, PO-coherent fragment)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable for PO-coherent concepts.*

Proof. Given C_0:

1. Compute $f(|C_0|) = D\,(1 + |\mathsf{cl}(C_0)|)$ with $D = 2^{2^{|\mathsf{cl}(C_0)|}}$.
2. Enumerate all two-tier quotients of size $\leq f(|C_0|)$.
3. For each, check validity: (V1)–(V4) for descriptors, PO-coherence, (V5) CC, (V_{safe}) phase-wise safety, (V6) off-chain path-consistency, (T4)–(T5) designated witnesses.
4. Accept iff any valid quotient is found.

Correctness: no false negatives (Theorem 7.1), no false positives (Theorem 8.3). All steps are computable. □

10 The PO gap

10.1 Why PO is different

For $R \in \{\mathsf{DR}, \mathsf{PP}\}$, backward forcing (Lemma 3.4) guarantees that a witness at a chain position past the exhaustion index M is R-related to *all* earlier chain positions. This yields all-phase type-safety (Lemma 4.6), enabling a constant-R kernel interface.

For $R = \mathsf{PPI}$, forward absorption ($\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$) ensures that the stabilized value of any PPI-witness *is* PPI, and Lemma 4.5 gives all-phase safety.

For $R = \mathsf{PO}$, *neither* property holds (Remark 3.5). The composition table permits PO at one chain position and a different relation (DR, PP, or PPI) at another.

10.2 A realizable PO-incoherent descriptor

Proposition 10.1. *There exists a satisfiable $\mathcal{ALCI}_{RCC5}$ concept C_0 whose model-extracted period descriptor $\mathsf{Desc} = (\tau_A, \tau_B)$ satisfies $\exists \mathsf{PO}.A \in \tau_A$ and $\forall \mathsf{PO}.\neg A \in \tau_B$, yet no element in the model has stabilized relation PO to the chain.*

Construction. Let the chain be $d_0(\tau_A)$ PP $d_1(\tau_B)$ PP $d_2(\tau_A)$ PP $d_3(\tau_B)$ PP $\cdots$. For each $k \geq 0$, create a witness w_k with $A \in \mathsf{tp}(w_k)$ and:

$$\rho(d_i, w_k) = \begin{cases} \mathsf{DR} & \text{if } i \leq 2k-1, \\ \mathsf{PO} & \text{if } i = 2k, \\ \mathsf{PPI} & \text{if } i \geq 2k+1. \end{cases}$$

The transitions $\mathsf{DR} \to \mathsf{PO} \to \mathsf{PPI}$ are valid by Lemma 3.6. Backward forcing gives DR at all positions before the PO event.

$\forall\mathsf{PO}.\neg A$ at τ_B-positions: At d_{2k+1} (type τ_B), the relation to w_k is PPI, to w_l for $l < k$ is PPI, and to w_l for $l > k$ is DR. No element is PO-related to d_{2k+1}, so $\forall\mathsf{PO}.\neg A$ is vacuously satisfied.

Mutual witness consistency: For $j < k$: $\rho(w_j, w_k) \in \mathsf{comp}(\mathsf{inv}(\rho(d_{2j+1}, w_j)), \rho(d_{2j+1}, w_k)) = \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$. So $\rho(w_j, w_k) = \mathsf{DR}$ for all pairs, which is composition-consistent.

Every w_k stabilizes to PPI. The concept is satisfiable, but no element has stabilized PO. □

Corollary 10.2. *The constant-interface quotient architecture is incomplete for PO-incoherent descriptors: a constant-*PO *witness at the kernel would be* PO*-related to τ_B-positions, violating $\forall\mathsf{PO}.\neg A$.*

Remark 10.3 (The DR analogue is impossible)**.** The analogous descriptor with $\exists\mathsf{DR}.A \in \tau_A$ and $\forall\mathsf{DR}.\neg A \in \tau_B$ is *unrealizable*: any DR-witness at a τ_A-position is DR at all earlier τ_B-positions (backward forcing), violating $\forall\mathsf{DR}.\neg A$. This is why the DR case has no gap.

10.3 The scope of the gap

The PO gap affects *only* descriptors that are PO-incoherent in the sense of Definition 5.4. Specifically, the gap arises only when:

1. Some phase τ_j contains $\exists\mathsf{PO}.C$.
2. Some other phase τ_k contains a $\forall\mathsf{PO}$-universal that is incompatible with *every* witness type satisfying C.

This is a relatively specific configuration. For the vast majority of $\mathcal{ALCI}_{RCC5}$ concepts—including all those without $\forall\mathsf{PO}$ restrictions, those where $\forall\mathsf{PO}$ universals are uniform across phases, and those where the witness type is compatible with all phases—the proof is complete.

11 Complexity

The quotient size bound is $D\,(1 + |\mathsf{cl}(C_0)|)$ with $D \leq 2^{2^{|\mathsf{cl}(C_0)|}}$. This is doubly exponential; we do not claim ExpTime membership. An ExpTime bound (matching the $\mathcal{ALCI}$ lower bound) would require showing the quotient size is singly exponential, which we leave open.

12 Discussion

12.1 The role of full RCC5 tractability

The soundness proof relies on Theorem 2.3: path-consistency decides satisfiability for the entire RCC5 algebra. Without it, we would need to verify global consistency directly. Full tractability collapses this gap: local consistency (checkable) implies global consistency (needed).

12.2 Why constant interfaces work for DR, PP, PPI

The backward forcing property ($\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$) is the key algebraic fact. When a DR-witness (resp. PP-witness) exists at chain position i_n, backward forcing makes it DR (resp. PP) at *all* earlier positions. Since all recurrent types appear at earlier positions, all-phase type-safety is automatic (Lemma 4.6).

For PPI, forward absorption ($\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$) plays the dual role: the stabilized value is always PPI, and Lemma 4.5 gives all-phase safety.

12.3 The algebraic root of the PO gap

PO is the unique base relation with *neither* backward forcing nor forward absorption. The composition entries $\mathsf{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ (not $\{\mathsf{PO}\}$) and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}$ (not $\{\mathsf{PO}\}$) are the algebraic root. Furthermore, the cycle consistency condition $\mathsf{PO} \in \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ fails, so even phase-indexed interfaces cannot make PO recur periodically along a PP-chain once interrupted by PPI.

12.4 Potential routes to closing the PO gap

1. **Proof that PO-incoherent descriptors are unrealizable.** If one could show that $\exists \mathsf{PO}.A \in \tau_j$ and $\forall \mathsf{PO}.\neg A \in \tau_k$ cannot coexist in any model's T_∞, the gap would vanish. Proposition 10.1 refutes this unless additional semantic constraints are identified.

2. **A non-constant quotient architecture.** Replacing constant kernel interfaces with per-position interfaces would handle PO but would make the quotient infinite (or require a different finiteness argument).

3. **Hybrid approach.** Use the two-tier quotient for DR/PP/PPI demands and a different mechanism (e.g., type elimination [2]) for PO demands.

12.5 Extension to $\mathcal{ALCI}_{RCC8}$

For RCC8, PP splits into TPP and NTPP. NTPP is transitive, but RCC8 is *not* fully tractable. The decidability of $\mathcal{ALCI}_{RCC8}$ remains open.

13 Conclusion

We have proved that concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable for the PO-coherent fragment. The proof rests on four pillars:

1. **Algebraic:** Backward forcing of DR and PP; forward absorption of PPI; self-absorption of PP.

2. **Semantic:** The phase-uniform type-safety lemma (Lemma 4.5) and the backward-forcing safety lemma (Lemma 4.6).

3. **Constructive:** One kernel per descriptor, yielding a computable size bound $D\,(1+|\mathsf{cl}(C_0)|)$ with no circularity.

4. **Constraint-theoretic:** Full RCC5 tractability bridges local consistency (V6) to global consistency (the model).

The chain-unfolding lift lemma (Lemma 8.1) ensures the finite V6 certificate governs the infinite unfolded model.

The decidability of $\mathcal{ALCI}_{RCC5}$ in full generality remains open due to the PO gap (Proposition 10.1). The PO obstacle is precisely localized: it arises only for PO-incoherent descriptors where different phases carry conflicting ∀PO universals. Whether this obstacle can be overcome within the two-tier framework or requires a fundamentally different approach is the key remaining question.

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